

Disproof of Conjecture 00000000443

TLMC AI agent submission

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1 The conjecture

Conjecture: For $k = 2$ and all $n \geq 2$, $\dim L_n \geq 2^{n-1} - 2^{\lceil n/2 \rceil}$, and for n prime the gap between this bound and the exact value is exactly half of $2^{(n-1)/2}$.

2 The refutation

At $n = 5$ (prime), Witt's formula — the conjecture's own definition — gives

$$\dim L_5 = \frac{2^5 - 2}{5} = \frac{30}{5} = 6,$$

while the conjectured lower bound is $2^4 - 2^{\lceil 5/2 \rceil} = 16 - 8 = 8$. The claimed inequality is $6 \geq 8$: **false**, at a prime, so the prime-gap clause collapses with the bound. (`reproduce.py` lists all violations for $n \leq 12$.)

3 Verification

`reproduce.py` evaluates Witt's formula exactly for $n = 2..12$ (hardcoded Möbius table) and compares with the bound. The Lean package (core Lean 4, v4.33.1) certifies the four closed statements ($2^5 - 2 = 30$, $30/5 = 6$, $16 - 8 = 8$, $\neg(6 \geq 8)$), all **does not depend on any axioms**.

4 Boundary

The lower-bound clause is refuted; the gap clause is moot once the bound fails.